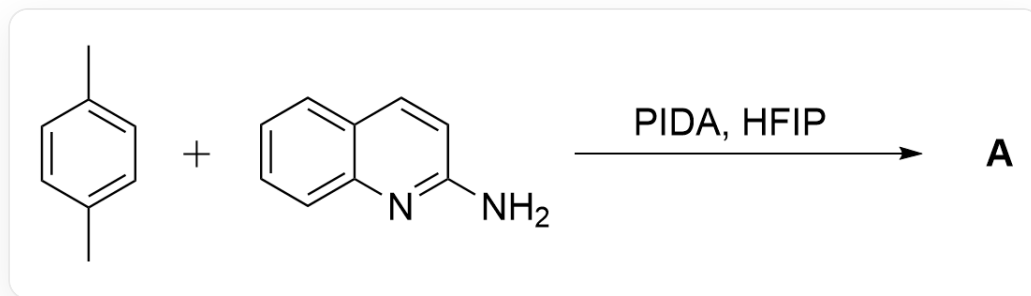


Question

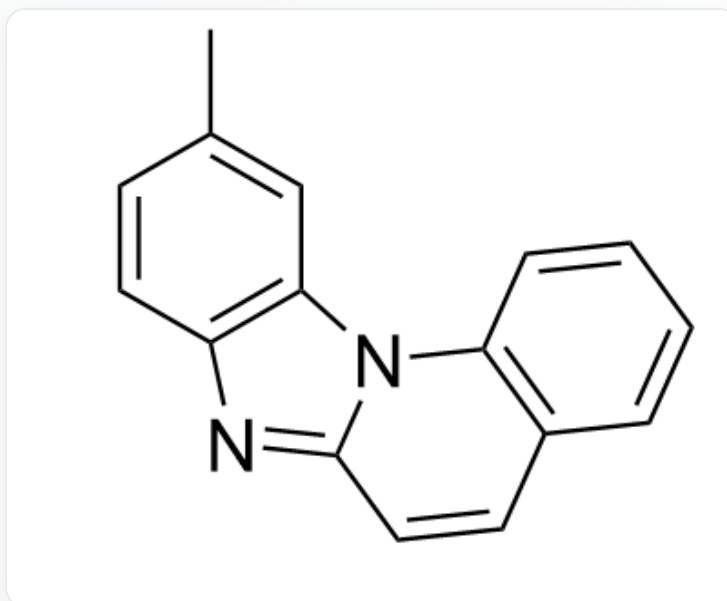


CC1=CC=C(C)C=C1.NC2=NC3=CC=CC=C3C=C2>[PIDA],[HFIP]>[A], A is the reaction product

In the process of generating **A**, PIDA first oxidizes p-xylene to a benzylic cation via a free radical mechanism. After being captured by aminoquinoline, it is further oxidized twice by PIDA to obtain product **A**. Please try to deduce the structure of product **A** based on this hint. (HFIP is hexafluoroisopropanol)

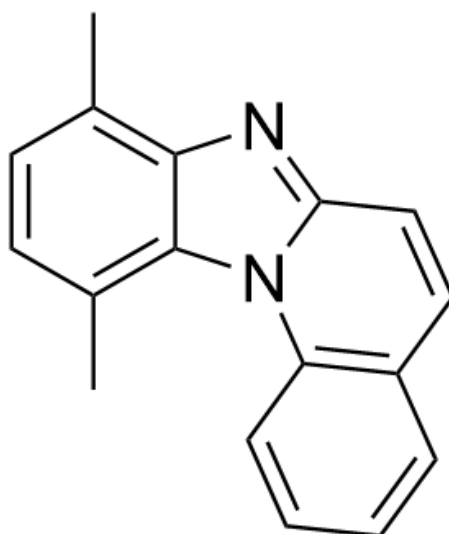
A. All other options are incorrect

B.



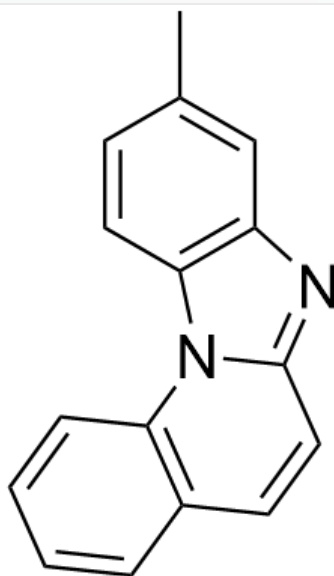
CC1=CC2=C(N=C3N2C4=C(C=CC=C4)C=C3)C=C1

C.



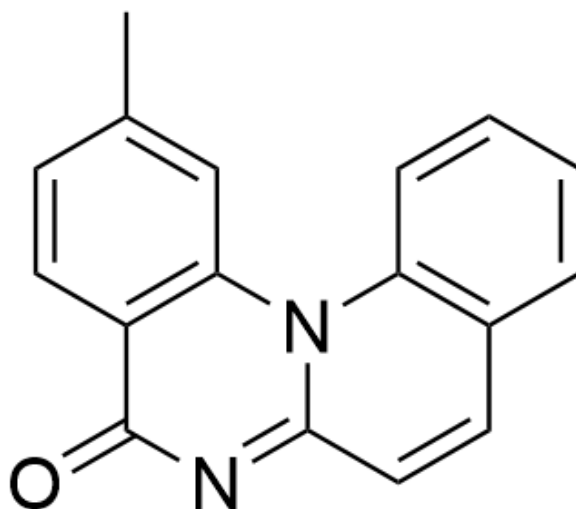
CC1=C2C(N(C(C=CC=C3)=C3C=C4)C4=N2)=C(C)C=C1

D.



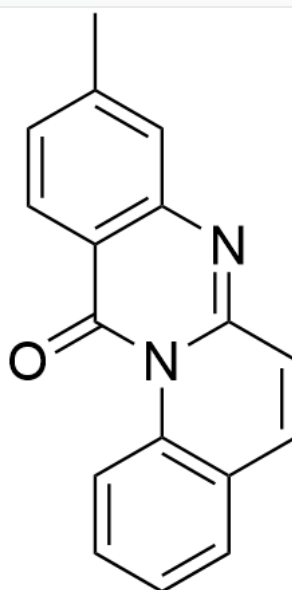
CC1=CC2=C(N(C(C=CC=C3)=C3C=C4)C4=N2)C=C1

E.



CC1=CC2=C(C(N=C3N2C4=C(C=CC=C4)C=C3)=O)C=C1

F.



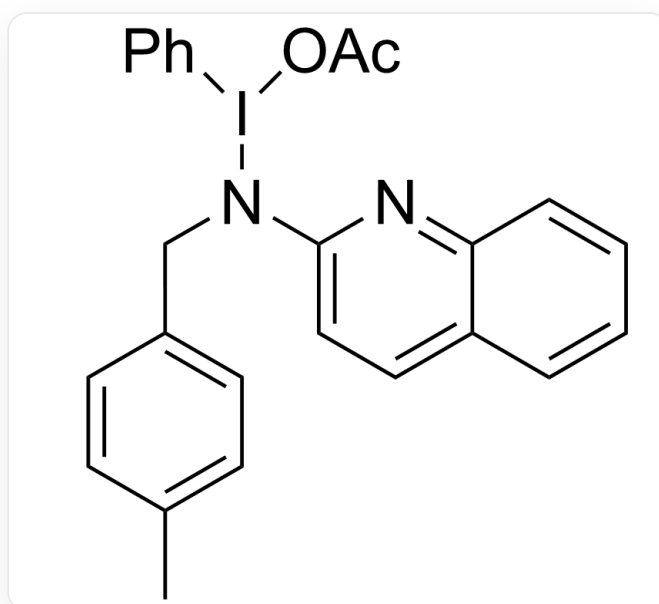
CC1=CC2=C(C(N(C(C=CC=C3)=C3C=C4)C4=N2)=O)C=C1

Answer

Correct Answer: **B**

Detailed Explanation

According to the question prompt, the benzylic cation intermediate generated in this reaction is first attacked by ammonia nucleophilically, and then the nitrogen atom combines with 1 molecule of oxidant PIDA to obtain the intermediate



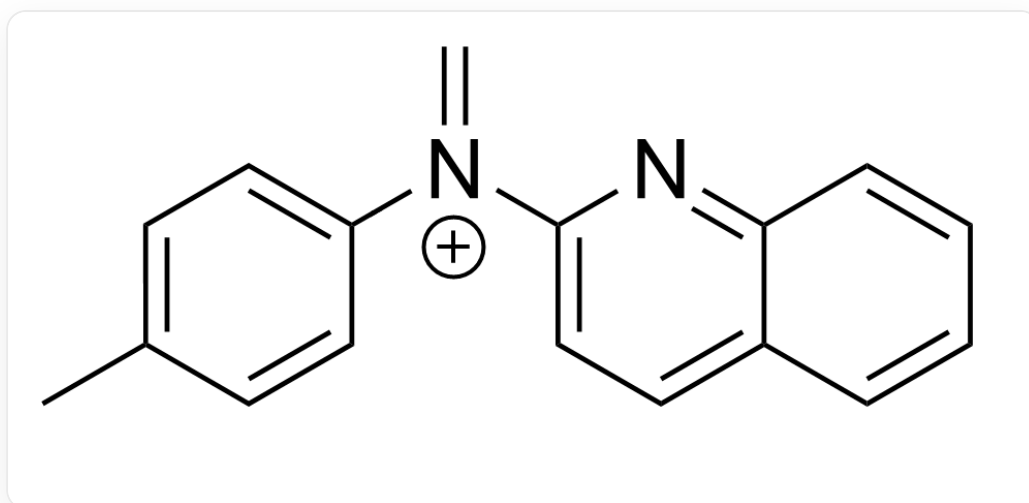
CC1=CC=C(CN(I(C2=CC=CC=C2)OC(C)=O)C3=CC=C(C=CC=C4)C4=N3)C=C1

CHECKPOINT

1 PTS

The first intermediate is CC1=CC=C(CN(I(C2=CC=CC=C2)OC(C)=O)C3=CC=C(C=CC=C4)C4=N3)C=C1

Then a phenyl migration reaction occurs to obtain the intermediate



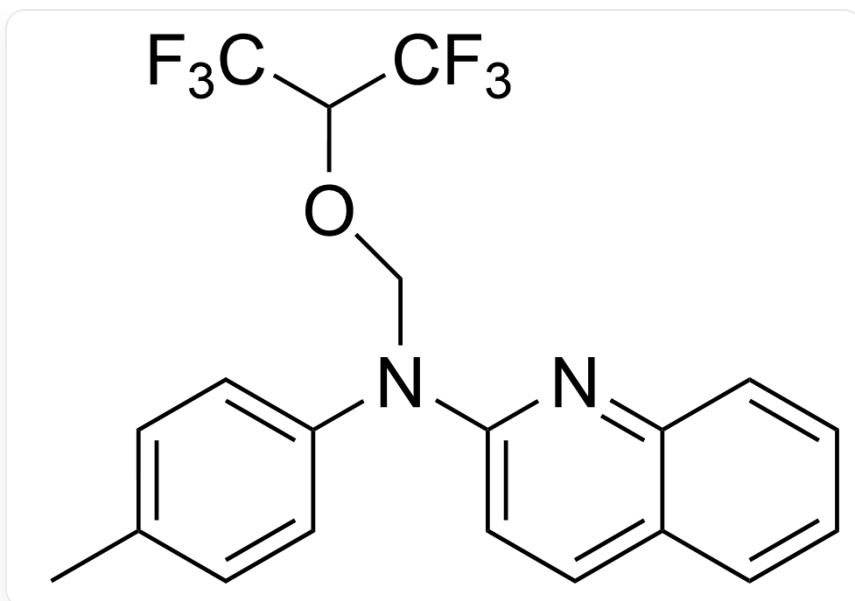
C=[N+](C1=CC=C(C=CC=C2)C2=N1)C3=CC=C(C)C=C3

CHECKPOINT

1 PTS

Then a phenyl migration reaction occurs to obtain the intermediate: C=[N+](C1=CC=C(C=CC=C2)C2=N1)C3=CC=C(C)C=C3

Subsequently, the intermediate is captured by the anion in the system



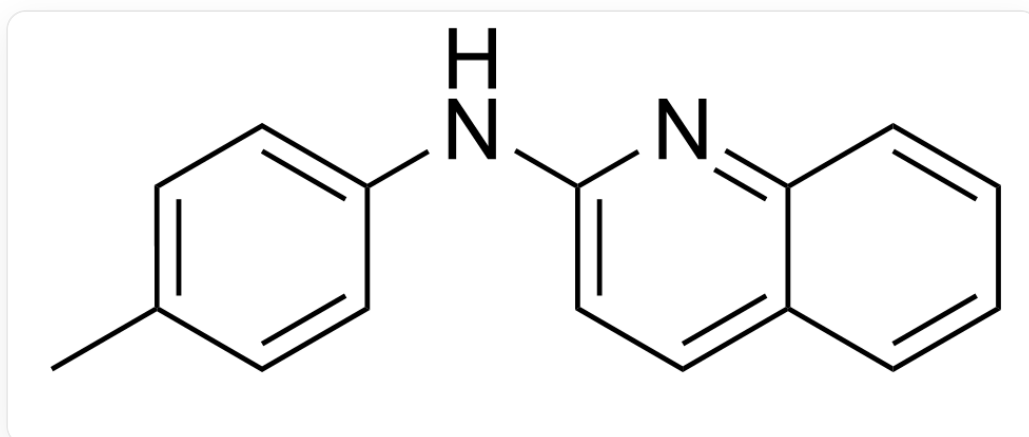
CC(C=C1)=CC=C1N(COC(C(F)(F)F)C(F)(F)F)C2=CC=C(C=CC=C3)C3=N2

CHECKPOINT

1 PTS

The third intermediate: CC(C=C1)=CC=C1N(COC(C(F)(F)F)C(F)(F)F)C2=CC=C(C=CC=C3)C3=N2

Then the secondary amine intermediate is reformed



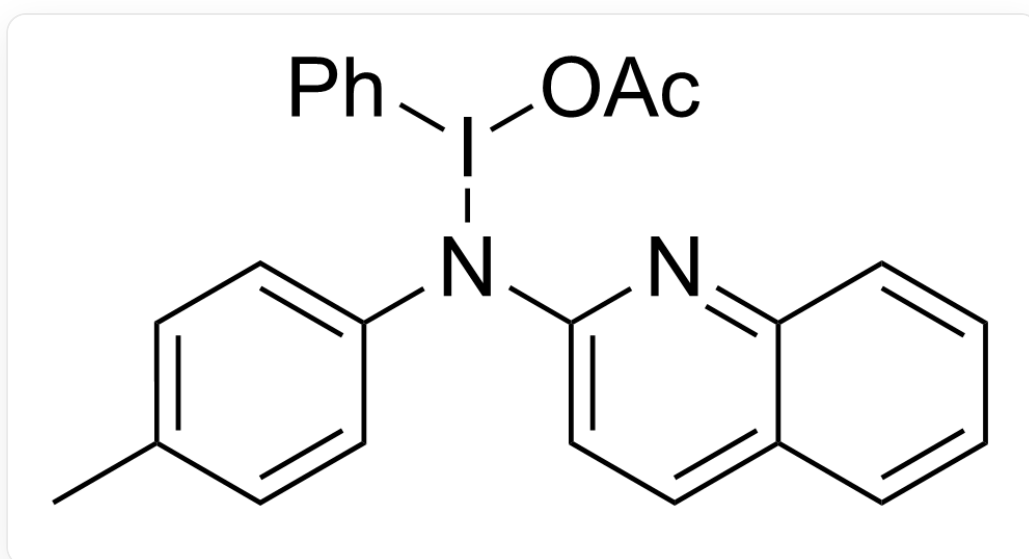
CC(C=C1)=CC=C1NC2=CC=C(C=CC=C3)C3=N2

CHECKPOINT

1 PTS

Secondary amine intermediate: CC(C=C1)=CC=C1NC2=CC=C(C=CC=C3)C3=N2

Subsequently, the nitrogen atom combines again with 1 molecule of oxidant PIDA



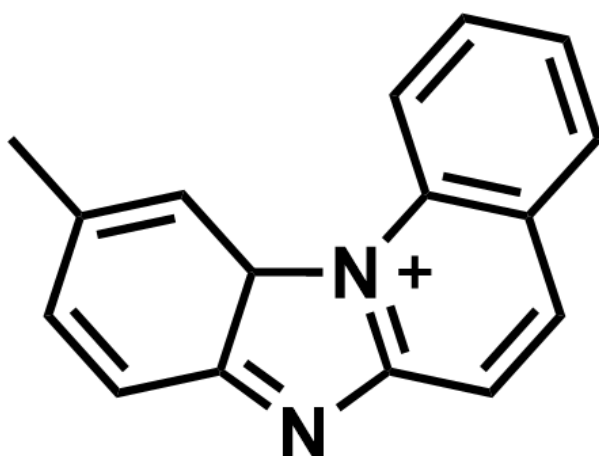
CC(C=C1)=CC=C1N(I(C2=CC=CC=C2)OC(C)=O)C3=CC=C(C=CC=C4)C4=N3

CHECKPOINT

1 PTS

The fifth intermediate: CC(C=C1)=CC=C1N(I(C2=CC=CC=C2)OC(C)=O)C3=CC=C(C=CC=C4)C4=N3

The nitrogen atom performs nucleophilic attack on the benzene ring, yielding the intermediate



CC1=CC2[N+]3=C(N=C2C=C1)C=CC4=C3C=CC=C4

CHECKPOINT

1 PTS

The sixth intermediate: CC1=CC2[N+]3=C(N=C2C=C1)C=CC4=C3C=CC=C4

Finally, the final product is obtained via deprotonation and aromatization